

3 Counting

3.1 Counting Lists

2. Airports are identified with 3-letter codes. For example, the Richmond, Virginia airport has the code *RIC*, and Portland, Oregon has *PDX*. How many different 3-letter codes are possible?

There are $26 \cdot 26 \cdot 26 = 17576$ possible 3-letter codes that are different.

4. Five cards are dealt off of a standard 52-card deck and lined up in a row. How many such line-ups are there in which all 5 cards are of the same color (i.e., all black or all red)?

From a 52-card deck, there are 26 of each color, from each half we have $26 \cdot 25 \cdot 24 \cdot 23 \cdot 22 = 7,893,600$ line-ups where all 5 cards are of that same particular color, taking into consideration it can be either of the two colors, then we have $2 \cdot 7,893,600 = 15,787,200$ line-ups where all 5 cards are of the same color.

6. Five cards are dealt off a standard 52-card deck and lined up in a row. How many such line-ups are there in which exactly one of the 5 cards is a queen?

We have 5 positions to place one of the 4 queens, that is $4 \cdot 5 = 20$ different ways, the remaining 4 positions can be any of the remaining 48 cards that are not a queen, therefore there are $4 \cdot 5 \cdot 48 \cdot 47 \cdot 46 \cdot 45 = 93,398,400$ line-ups with such arrangement.

8. This problem concerns lists made from the symbols *A, B, C, D, E*.

a) How many such length-5 lists have at least one letter repeated?

b) How many such length-6 lists have at least one letter repeated?

- a) We have 5^5 different possible lists and $5!$ which don't repeat, thus we have $5^5 - 5! = 3005$ length-5 lists with at least one letter repeated.
- b) There are more positions than symbols, thus all possible length-6 lists have at least one letter repeated, so there are $5^6 = 15,625$ such lists which have at least one letter repeated.

10. This problem concerns lists made from the letters $A, B, C, D, E, F, G, H, I, J$.

- a) How many length-5 lists can be made from these letters if repetition is not allowed and the list must begin with a vowel?
- b) How many length-5 lists can be made from these letters if repetition is not allowed and the list must begin and end with a vowel?
- c) How many length-5 lists can be made from these letters if repetition is not allowed and the list must contain exactly one A?

- a) There are 3 vowels for the first position, the remaining 4 positions are filled from the other 9 without repetition, so there are $3 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 9,072$ such lists.
- b) There are 3 choices of vowel for the first position and 2 for the last, and the middle 3 positions are filled from the remaining 8 letters, so there are $3 \cdot 2 \cdot 8 \cdot 7 \cdot 6 = 2,016$ such lists.
- c) The single A can go in any of the 5 positions, and the remaining 4 positions are filled from the 9 letters that are not A , so there are $5 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 15,120$ such lists.

12. Consider the lists of length six made with the symbols P, R, O, F, S , where repetition is allowed. (For example, the following is such a list: (P, R, O, O, F, S) .) How many such lists can be made if the list must end in an S and the symbol O is used more than once?

The last position is fixed as S , leaving the first 5 positions free, so there are $5^5 = 3125$ lists ending in S . From these we subtract the ones where O appears fewer than two times: $4^5 = 1024$ lists with no O , and $5 \cdot 4^4 = 1280$ lists with exactly one O (5 positions for the O , the other 4 filled from the 4 remaining symbols). Thus there are $5^5 - 4^5 - 5 \cdot 4^4 = 821$ such lists.